

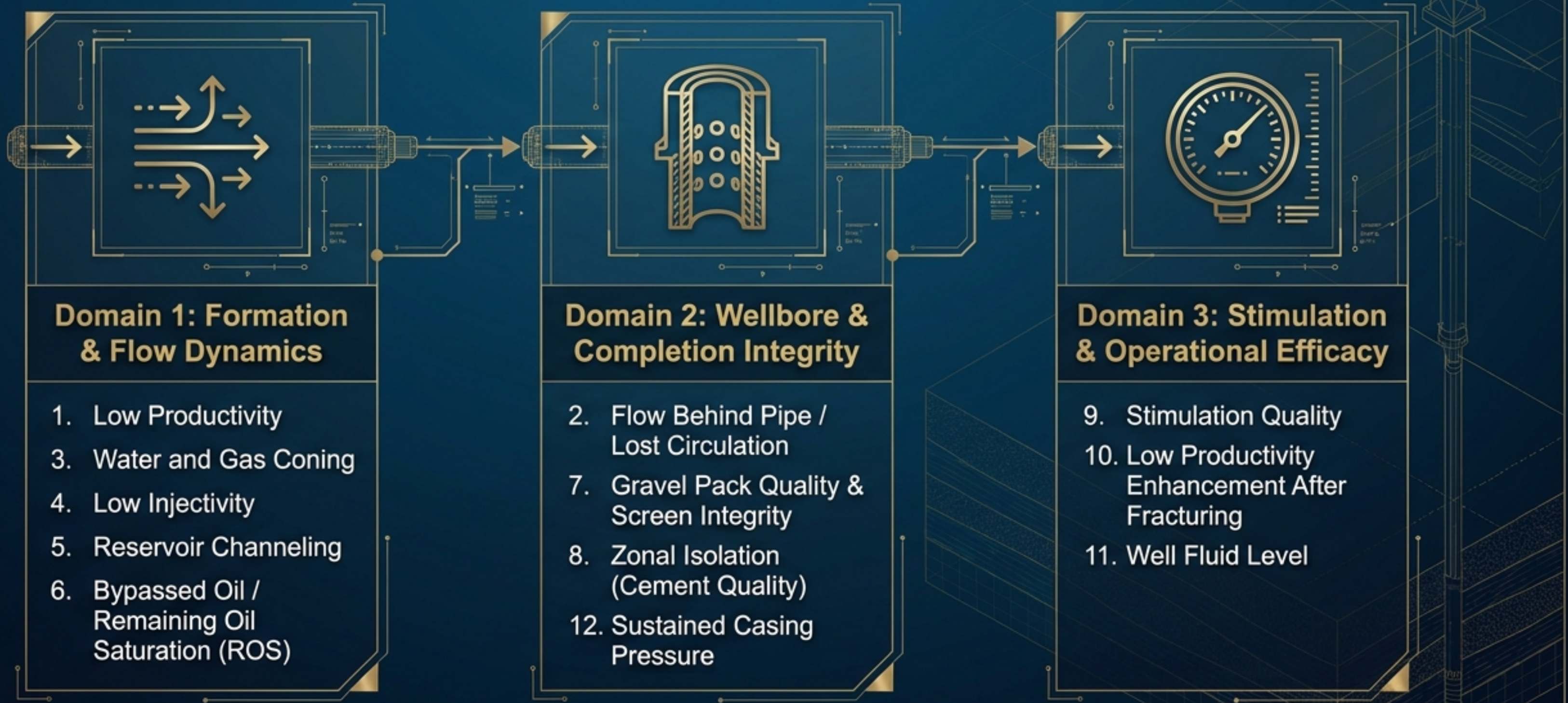
Reservoir Problem-Based Diagnostic Tools Selection

A Comprehensive Methodological Framework
for Subsurface Intervention

Prepared by Eng. sajjad k.Hano



The Diagnostic Taxonomy



Productivity & Injectivity Impairment

Problem 1: Low Productivity	
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Etiology	Low permeability, Plugged perforations, Near-wellbore damage, Channeling/cross flow, Fill over perforations, Scale/paraffin/asphaltene.
Diagnostic Tools	Pressure transient tests, Temp and spinner survey, Slickline gauge ring runs (to tag fill), Bailer samples of solids.

PRODUCTION



INJECTION

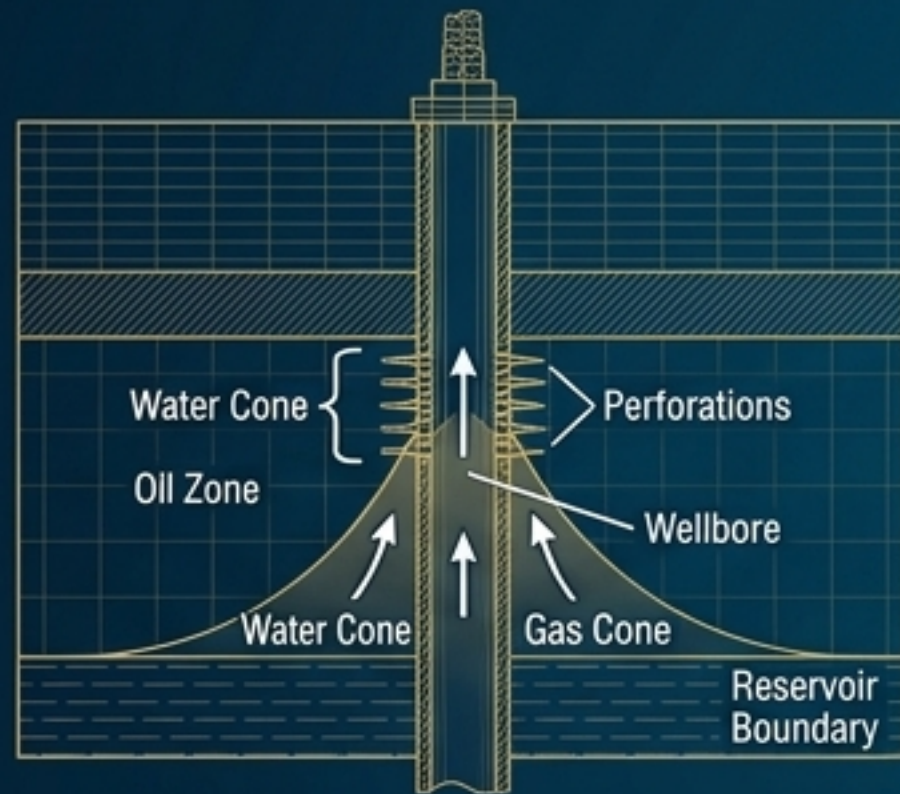
Problem 4: Low Injectivity	
Problem 4: Low Injectivity	
Etiology	Mechanically similar to low productivity.
Diagnostic Tools	Temp and RA tracer or spinner, Caliper log (ID restrictions), Pressure buildup or drawdown tests, Injection water quality, Sand fill bailers.

Clinical Notes (Constraints)

- Compare production profiles with cores and openhole logs to distinguish between low permeability and plugged perforations.
- Caliper logs can isolate tubing restrictions. Crossflow during shut-in is a common source of sand fill.

Unwanted Fluid Migration: Coning vs. Channeling

Problem 3: Water/Gas Coning (Vertical)

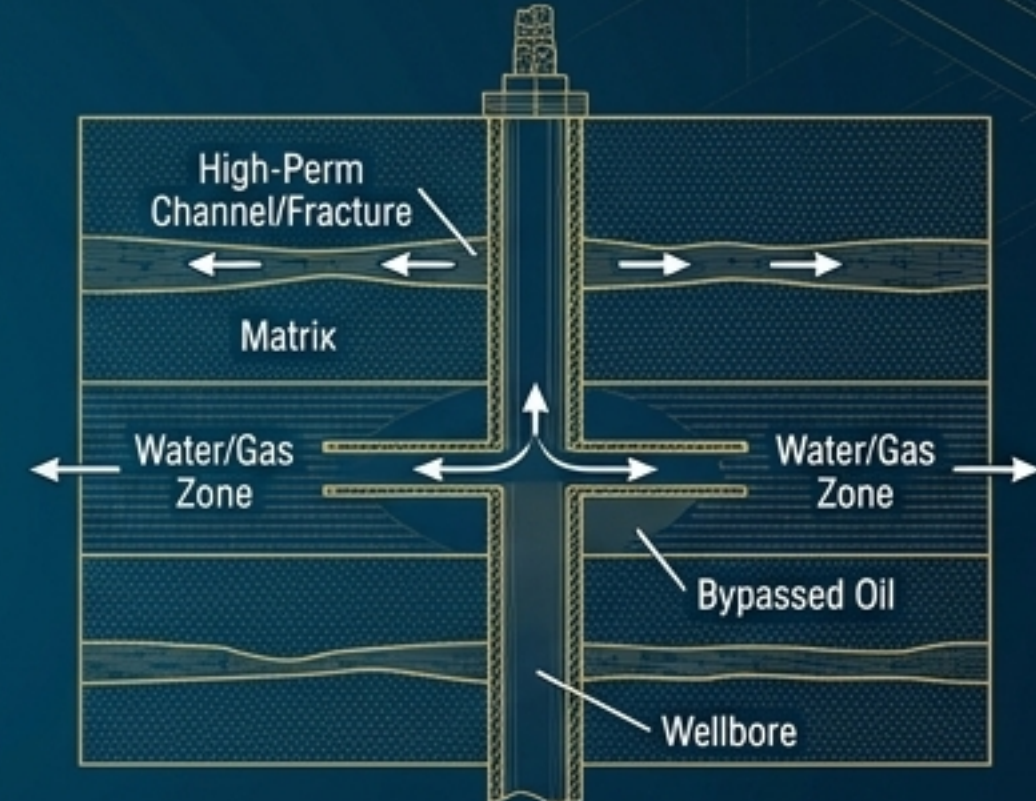


Reasons: Significant mobility differences, High production rates.

Tools: Production logs at multiple rates, Noise log (to eliminate channeling), Pulsed neutron log or C/O log (under shut-in and flowing conditions).

Constraints: Difficult to assess. Requires an elimination process using multiple tools. Exploit the rate sensitivity of coning behavior.

Problem 5: Reservoir Channeling (Lateral)



Reasons: High mobility ratios, Large transmissibility contrasts, Fractures acting as conduits.

Tools: Production logs or RA tracer, Pulse tests or interwell tracer tests, Oxygen activation log.

Constraints: Reservoir channeling may be specifically identified by pulse or interwell tracer tests.

Bypassed Oil & Remaining Oil Saturation (ROS)

Problem 6 Diagnostic Profile

Etiology

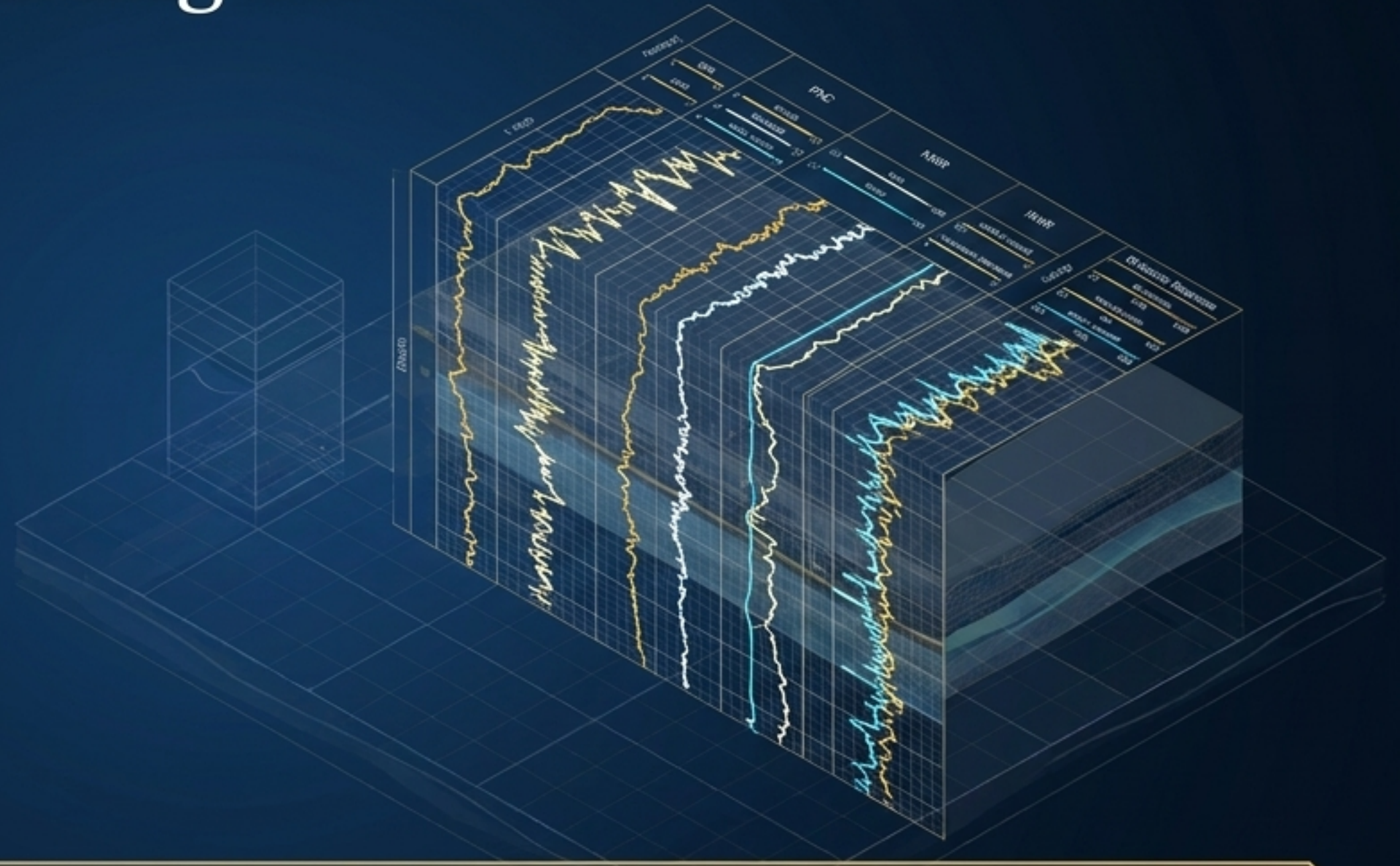
Stacked sand with transmissibility contrasts.

Instrumentation Suite

1. Time-lapse PNC & C/O log
2. Log-inject-log (LIL) with PNC
3. NMR MnCl_2 doping or LIL
4. Dielectric logging
5. Single well tracer survey

Operational Constraints

- PNC logs utilize log-inject-log (LIL) technique with fresh and salt-water injection.
- NMR requires drilling zone of interest with MnCl_2 .
- NML-LIL requires underreaming wellbore.
- Dielectric logs determine S_0 in invaded zone.
- Strict control of overbalance is required to prevent oil stripping.



Extraneous Flow & Lost Circulation Zones

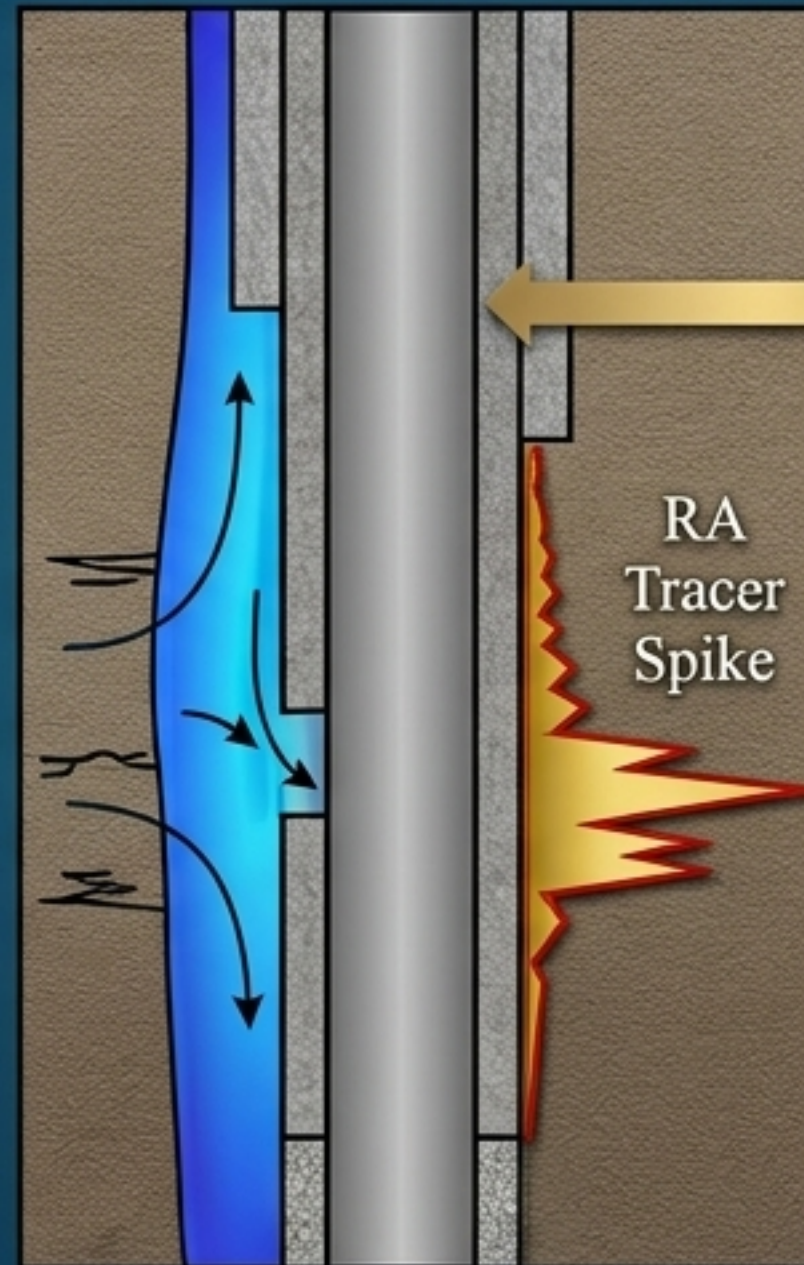
Problem 2: Flow Behind Pipe

Etiology

- Difference in pressure between zones.
- Lack of hydraulic isolation behind pipe.
- Encountering natural fracture systems.

Diagnostic Tools

1. Temperature logs
2. Radioactive (RA) tracer logs
3. Oxygen activation
4. Image logs



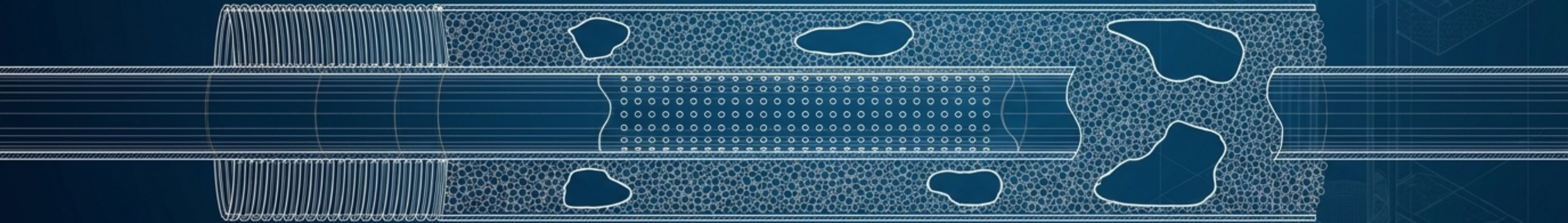
Key Diagnostic Signatures

Thermal: Look for a cool anomaly on a temperature log.

Radiological: A slug of injected tracer can be logged precisely with a gamma-ray tool.

Completion Degradation: Gravel Packs & Screens

Problem 7: Gravel pack quality and screen integrity



Etiology & Tools

Etiology: Poor packing, Screen failure.

Diagnostic Tools:

- Unfocused gamma ray density
- Ultrasonic sand detection
- PLT w/ R/A tagged gravel and carrier fluids

Clinical Constraints & Operational Alerts

- Warning: No logging tools monitor screen condition proactively; failures are detected after the fact.
- Gamma-ray density can locate voids in the pack.
- PLT can identify hot spots.
- Must prevent flow concentration exceeding erosional velocity.
- Water quality monitoring identifies scaling tendency and pitting.

Zonal Isolation & Cement Quality

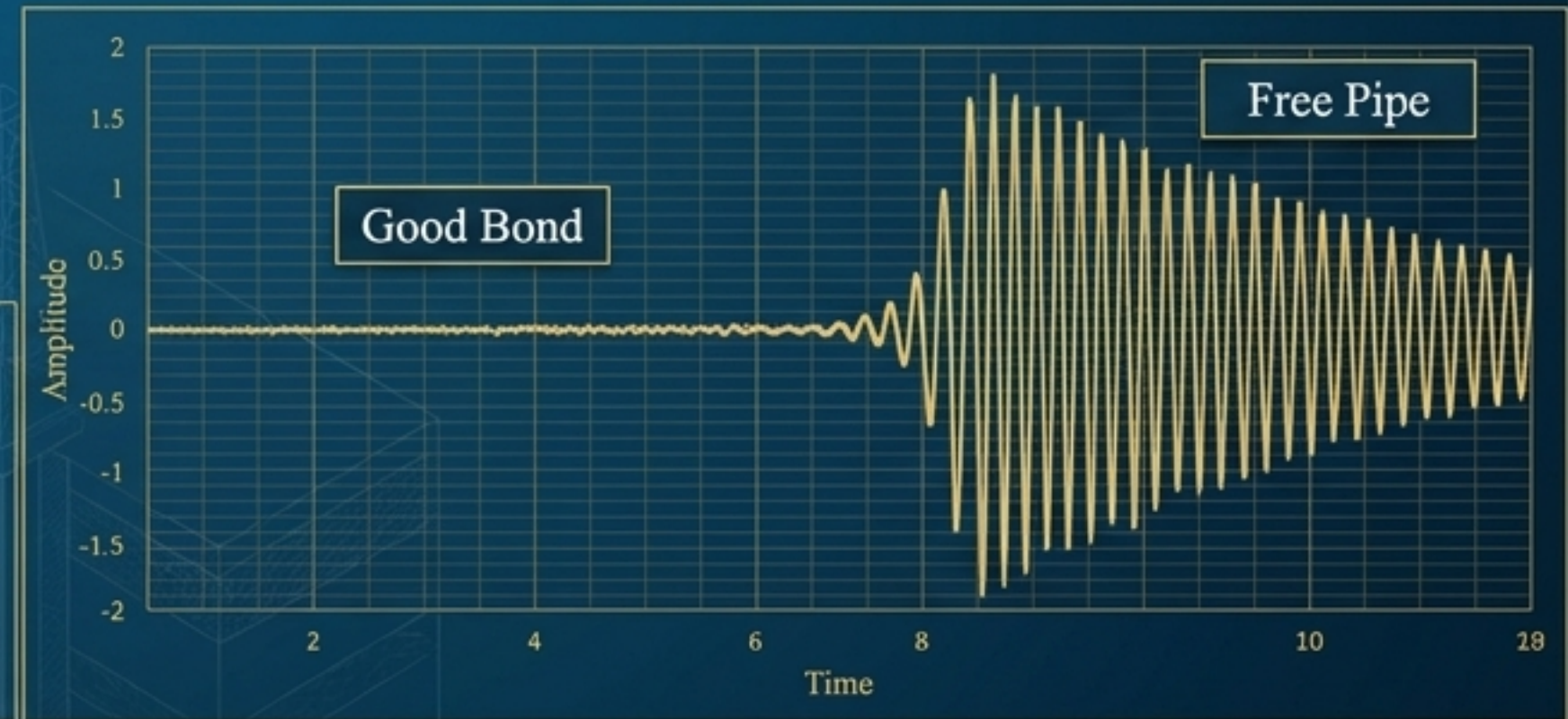
Problem 8: Cement Bond Integrity

Etiology:

- Cement bond not achieved.
- Cracks develop in cement.

Diagnostic Instrumentation:

1. Cement Bond Log (CBL)
2. Ultra sonic pulse echo
3. Temperature
4. Oxygen activation with PNC log
5. Ultrasonic leak detection tools



Execution Timing

- Cement evaluation logs provide a crucial record of zonal isolation before the well is on production.
- They should strictly be run after the primary cement job, before perforation or completion equipment is run.

Stimulation Efficacy & Post-Frac Enhancement

Problem 9: Stimulation Quality

Etiology: Poor fracture conductivity; determining that wellbore damage was not the cause of poor performance.

Tools: Pressure transient tests, Production logs.

Constraint: Accurate diagnosis requires establishing a baseline; pre- and post-tests must be conducted.

Problem 10: Low Productivity Enhancement



Etiology: Out of zone growth; incomplete zone coverage resulting in partial penetration effects.

Tools: Tracer survey (multi-isotope spectroscopy), Microseismic.

Constraint: Microseismic limitations—fracture growth oblique to the wellbore can be missed entirely.

Wellbore Hydraulics: Fluid Levels

Problem 11: Well Fluid Level Assessment

Etiology:
Poor pump performance.

Primary Diagnostic Tool:
Acoustic well sounding.



Diagnostic Constraint (Clinical Note)

Signal distortion: The presence of an emulsion or foamy gas/oil interface can severely scatter the acoustic signal, creating significant difficulties in determining the exact fluid level.

Pressure Anomalies: Sustained Casing Pressure

Problem 12: Sustained Casing Pressure (SCP)

Etiology:

- Packer or tubing leak.
- Poor primary cement job.
- Cement cracking or microannulus formation.

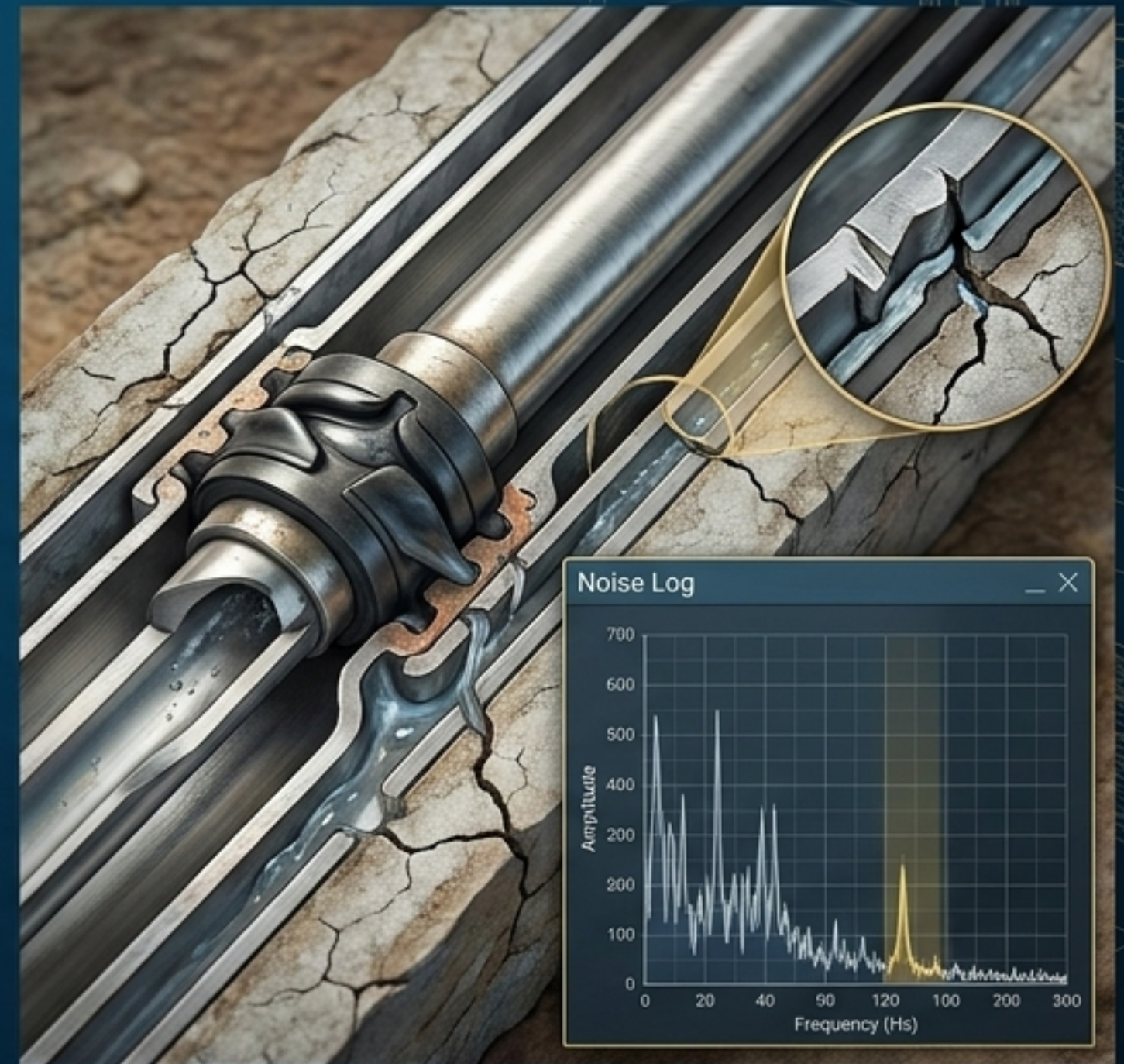
Diagnostic Tools:

- Ultrasonic detection
- Noise log

Diagnostic Challenge

Low leak rates are exceptionally difficult to diagnose definitively, often requiring high-resolution tools or extended monitoring periods to isolate the source.

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The Master Diagnostic Matrix

Cross-referencing technical etiology with optimal instrumentation

	Flow/Production Logs	Thermal/Tracer	Pressure Transient	Acoustic/Ultrasonic	Advanced Nuclear (PNC/NMR)
P1: Productivity			✓		
P2: Flow Behind Pipe		✓		✓	
P3: Coning	✓		✓		
P4: Injectivity	✓	✓	✓		
P5: Channeling		✓		✓	
P6: Bypassed Oil/ROS	✓				✓
P7: Gravel Pack/Screens	✓			✓	
P8: Zonal Isolation		✓		✓	✓
P9: Stimulation Quality	✓		✓	✓	
P10: Post-Frac Enh.	✓				✓
P11: Fluid Level			✓	✓	
P12: Sustained Casing Pres.	✓	✓		✓	